

Fine-Tuning a Foundation Music Model for Carnatic Audio Continuation

LoRA adaptation of MusicGen-Small on ~96 hours of authentic Carnatic recordings, evaluated through a hybrid quantitative + perceptual framework.

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The Problem

Foundation music models are systematically biased toward Western traditions

What's broken

Modern foundation music models (MusicGen, MusicLM, AudioLDM) are pretrained on Western-dominated corpora.

Carnatic music, raga grammar, microtonal gamakas, cyclic talas, sits outside that distribution.

Why current solutions don't fit

Text-to-audio adaptation requires prompt-labelled corpora. For Carnatic, labels are scarce, subjective, and culturally hard to standardize (Ravikumar, 2024).

~96 hrs

Cleaned authentic Carnatic audio
(CompMusic, UPF)

~0.5%

Trainable parameters via LoRA —
pretrained core preserved

32 listeners

Hybrid perceptual study, Holm-
corrected significance

Research Questions

Three concrete, testable claims — one quantitative, one acoustic, one perceptual

RQ1

Trainability

Can MusicGen-Small be adapted to Carnatic with a small fraction of trainable parameters (LoRA) while producing measurable improvement in token-level cross-entropy loss?

RQ2

*Quantitative
continuity*

Does LoRA fine-tuning improve acoustic continuity at the prompt-to-continuation boundary on Mel, MFCC, Chroma, and Onset cosine distances?

RQ3

*Perceived
authenticity*

Do human listeners — both Carnatic-familiar and unfamiliar — prefer the fine-tuned continuations on musicality, Carnatic authenticity, and continuity, with statistically significant margins after multiple-comparison correction?

Design Decision: Audio-to-Audio, Not Text-to-Audio

Text-to-Audio — REJECTED

Examples: Mustango, Instruct-MusicGen, ChatMusician

- Requires large paired prompt-audio datasets.
- Carnatic prompt-labelled corpora don't exist at scale; cultural labelling is subjective (Ravikumar, 2024).
- Even with labels, prompt-alignment evaluation imports its own bias.
- T5 text encoder dominates the input path — wrong leverage point for cultural adaptation.

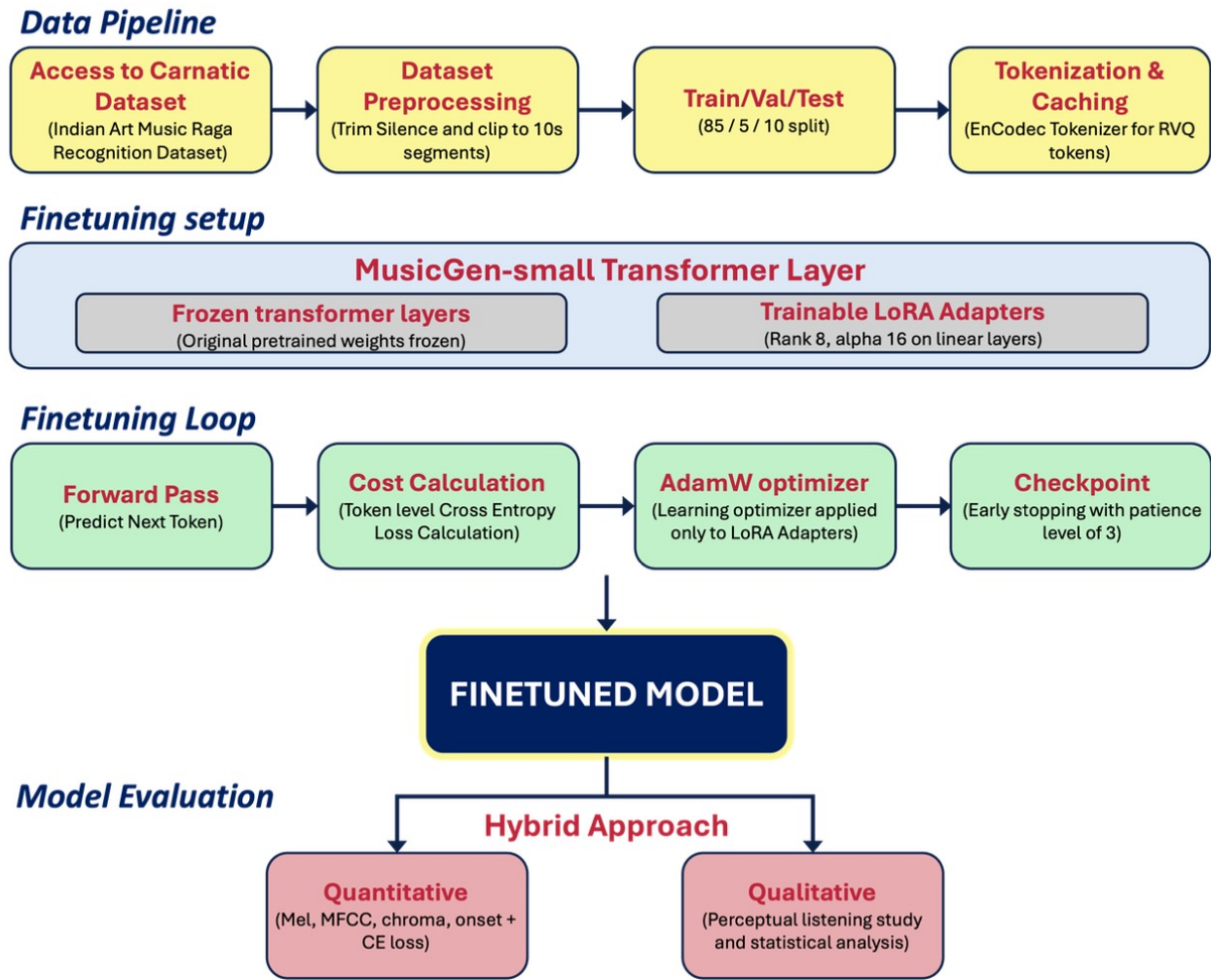
Audio-to-Audio Continuation — CHOSEN

MusicGen's generate_continuation API; no text supervision

- Uses pure audio — culturally rich datasets are usable as-is.
- Adapts the LM directly on EnCodec RVQ tokens — the actual generative bottleneck.
- Continuation framing yields natural boundary-anchored evaluation (prompt ↔ continuation seam).
- Recipe is replayable for any underrepresented tradition that has audio but lacks labels.

Methodology Overview

Ten sequential stages, every stage emits an inspectable artifact — re-runnable independently



Reproducibility: fixed seeds (split = 42), per-stage CSVs, AudioCraft Apple-Silicon fork pinned in `initial_environment_setup.md`

Dataset Choice

Three candidates considered; only one cleared the licensing × authenticity × structure bar

Source	License	Authenticity	Structure	Verdict
YouTube collections	Unclear / mixed	Variable	Inconsistent metadata	Rejected
Kaggle dumps	Uncertain provenance	Unverified	Ad hoc	Rejected
Saraga (CompMusic)	Research	High	Annotation-heavy, smaller	Set aside
Indian Art Music Raga Recognition Dataset (Serrà et al., 2016)	Zenodo, research-only	Curated, authentic	Artist / raga directory	SELECTED

Why this matters: An research project lives or dies on data provenance. The same dataset has been used for vector-space raga recognition (Gulati et al., 2016) but never for foundation-model domain adaptation.

Preprocessing

- 1 Resample to 32 kHz mono**
Native EnCodec / MusicGen input rate; avoids resampling artifacts inside the model.
- 2 Trim leading/trailing silence < 0.01**
Eliminates dead headroom that would dominate boundary metrics; threshold low enough to preserve quiet onsets.
- 3 Non-overlapping 10-s clips (320,000 samples)**
Matches MusicGen training window; non-overlapping prevents within-recording leakage.
- 4 RMS ≥ 0.005 , near-zero ratio ≤ 0.30 , non-silent ratio ≥ 0.60**
Three orthogonal heuristics: rejects near-silent, silence-heavy, and sparse-active clips respectively.
- 5 Recording-level train/val/test split (85/5/10), seed=42**
Critical — segment-level splits leak artist/raga across folds. Recording-level is the only defensible choice.

Preprocessing outcome

426

source recordings

39,162

10-second segments

34,400

accepted (12.1% rejected)

~95.6 hrs

effective training corpus

361 / 22 / 43

train / val / test recordings

Model Choice: Why MusicGen-Small?

Why Small is sufficient for the claim

- Goal is to test whether LoRA can shift a foundation model.
- Smaller models surface adaptation effects more cleanly; less pretrained inertia to overcome.
- Recipe transfers directly to Medium / Large on accelerator hardware.
- Demonstrates the resource floor: cultural adaptation is feasible on commodity hardware which is a real contribution.

Adaptation Method: LoRA vs Alternatives

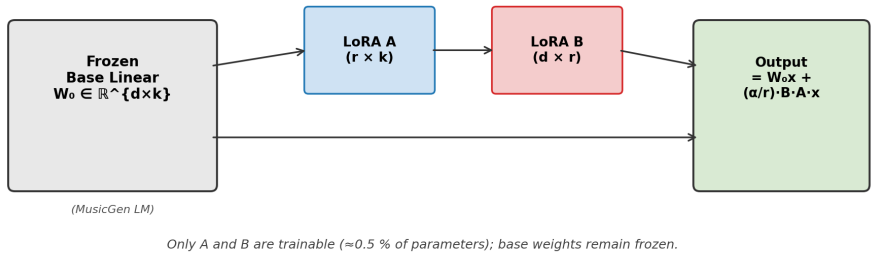
Method	Trainable params	Forgetting risk	Compute fit (MPS)	Verdict
Full fine-tuning	100%	High — catastrophic forgetting	Infeasible	Rejected
Adapter modules (insert blocks)	~3-5%	Medium — alters architecture	Marginal	Set aside
Prompt / prefix tuning	<0.1%	Low	Fits	Too weak for distribution shift
LoRA (chosen)	~0.5%	Low — base weights frozen	Fits on MPS	SELECTED

The LoRA decision

- LoRA injects two low-rank matrices $A \in \mathbb{R}^{d \times k}$ and $B \in \mathbb{R}^{k \times d}$ into every Linear layer of the MusicGen LM. Base weight W_0 stays frozen; the layer outputs $y = W_0 x + (\alpha/r) \cdot B \cdot A \cdot x$. A is Kaiming-initialised, B is zero.
- The adapter contributes nothing at step 0 and only deviates from the pretrained behaviour as it learns.

LoRA Hyperparameters

Figure: LoRA Adapter Injection on Each Linear Layer of MusicGen LM (rank $r=8$, $\alpha=16$)



LoRA injection - applied recursively to every nn.Linear in the LM transformer (Q, K, V, output projections + MLP layers). condition_provider modules are deliberately excluded because this study performs unconditional audio continuation.

Hyperparameter	Value	Rationale
LoRA rank (r)	8	Standard low-rank choice; $\approx 0.5\%$ trainable params
LoRA alpha (α)	16	Effective scaling $\alpha/r = 2$ - standard for PEFT
LoRA dropout	0.05	Light regularization only on the adapter path
Optimizer	AdamW	Decoupled weight decay; standard for transformer FT
Learning rate	1e-4	Conservative - preserve pretrained knowledge
Weight decay	1e-4	Regularization against runaway updates
Batch / grad accumulation	1 $\times$ 8	Effective batch 8; MPS / RAM compatible
Gradient clipping	max norm 1.0	Stability against early-training spikes
Epochs	10, patience 3	Early-stop on val CE; best epoch selected

Training Loop: Engineering Choices

EnCodec token caching

- Every accepted 10-s segment encoded once and cached as a torch payload.
- Removes redundant tokenization cost from every epoch.

Impact: Made the full pipeline runnable on MPS without saturating GPU memory.

Unconditional mode via dummy conditioning

- Built a single empty ConditioningAttributes on CPU per batch instead of invoking the T5 text encoder.
- The condition_provider modules are excluded from LoRA wrapping entirely.

Impact: Aligned the training signal with the actual inference task: pure audio continuation.

Loss: token-level cross-entropy via compute_predictions

- Standard MusicGen objective on the masked LM output against cached EnCodec RVQ codes.
- Validation CE checked every epoch; best-epoch LoRA checkpoint extracted.

Impact: Tiny on-disk footprint saving only LoRA matrices (~0.5% of base size).

Wrapping decision:

- LoRA injection restricted to nn.Linear modules only (Q, K, V, output projection, and the two MLP linears per block).
- Linear layers carry almost all the trainable parameters in a transformer, so adaptation here is the highest-leverage surface.

Evaluation Strategy: Why Hybrid?

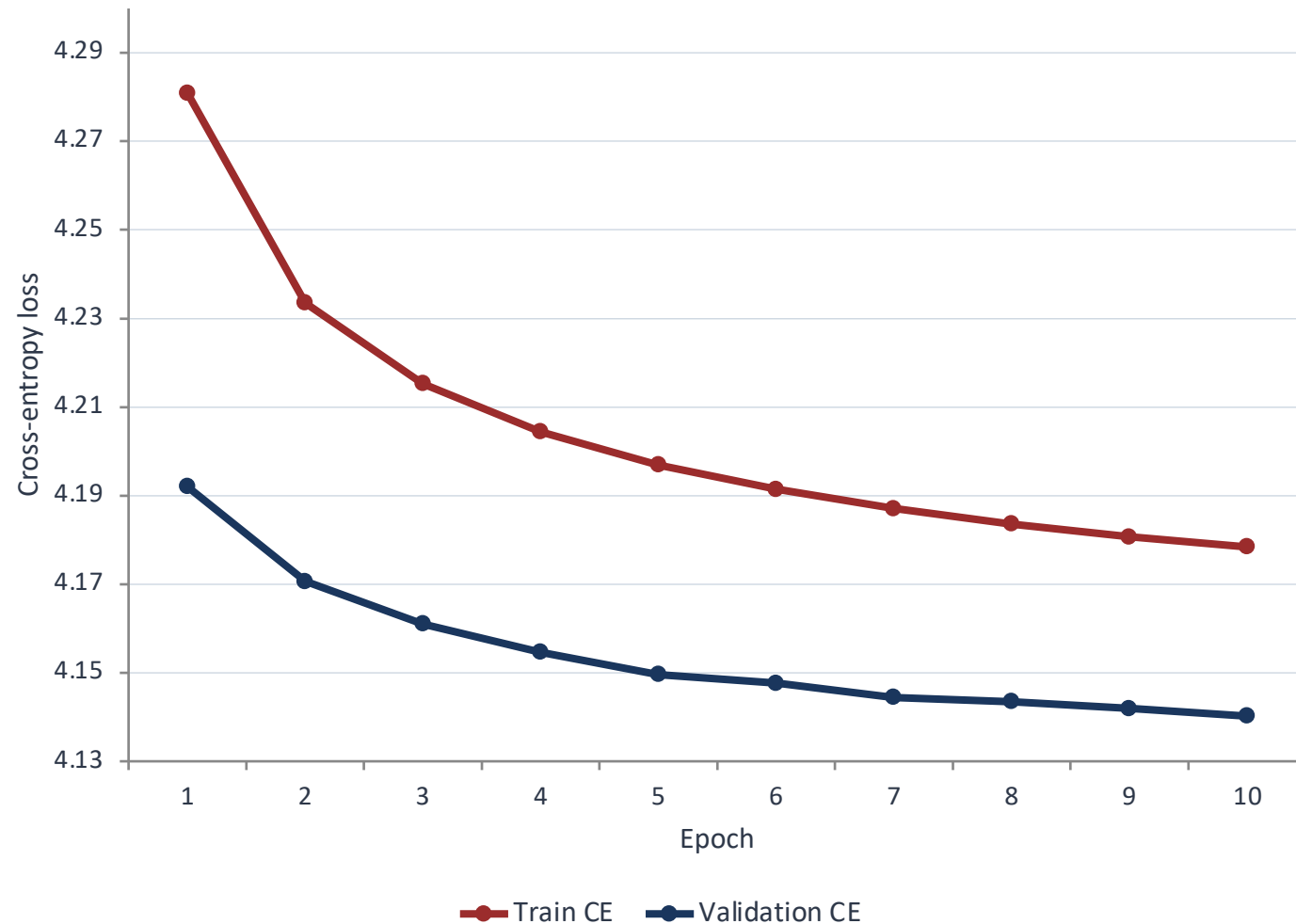
Music quality is subjective and multi-faceted: neither objective nor perceptual evaluation alone is sufficient

Layer 1 RQ1	Training dynamics Cross-entropy loss and perplexity per epoch. Confirms the model is learning the Carnatic distribution and not diverging.
Layer 2 RQ2	Boundary-continuation metrics Cosine distance on Mel, MFCC, Chroma, Onset features at the prompt → continuation seam. Per-clip acoustic continuity.
Layer 3 RQ3	Human listening study Likert MOS (musicality, continuity, Carnatic authenticity) + forced-choice A/B preference, with Holm-corrected one-sided binomial tests.

Lerch et al. (2025) — survey on music-generation evaluation — explicitly advocates for hybrid frameworks; statistical similarity alone does not guarantee perceptual quality (Guo et al., 2023).

Training Dynamics — RQ1

10-epoch LoRA run: monotonic decrease, no divergence, no overfitting



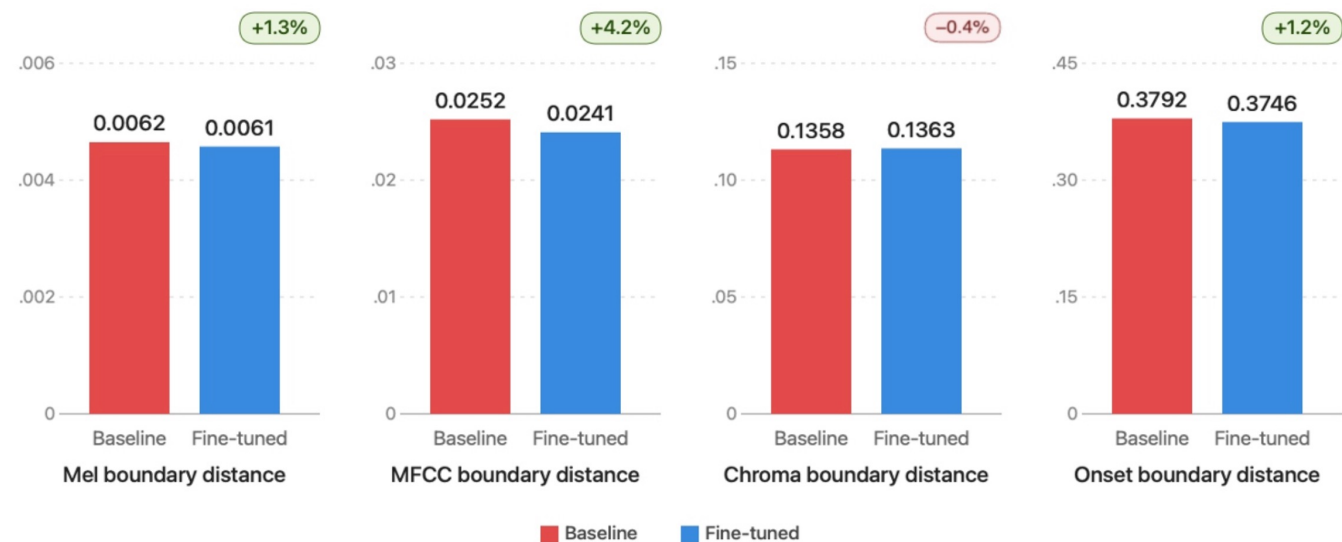
Two takeaways

1. Validation CE drops $4.192 \rightarrow 4.140$ (≈ 0.052 nats; perplexity $66.18 \rightarrow 62.84$) while only **$\sim 0.5\%$** of parameters are trainable. The pretrained foundation is preserved.
2. The curve is still trending downward at epoch 10. Longer training or larger LoRA rank would likely yield further improvement

Verdict on RQ1: answered affirmatively.

Boundary-Continuation Metrics — RQ2

Mean cosine distance at the prompt → continuation seam (lower = smoother)



Carnatic relevance of each feature

Mel	Vocal / instrumental tone continuity
MFCC	Bowing & attack - most sensitive to articulation
Chroma	12-bin grid can't represent microtones / gamakas
Onset	Proxy for tala-style consistency

Honest caveat — per-clip wins ≈ 50%

Means improve consistently, but per-clip the fine-tuned model strictly beats baseline on only 49–51% of clips. 1-second cosine distances are noisy per-sample estimators. This is exactly why a perceptual study was necessary.

Listening Study: Design Choices

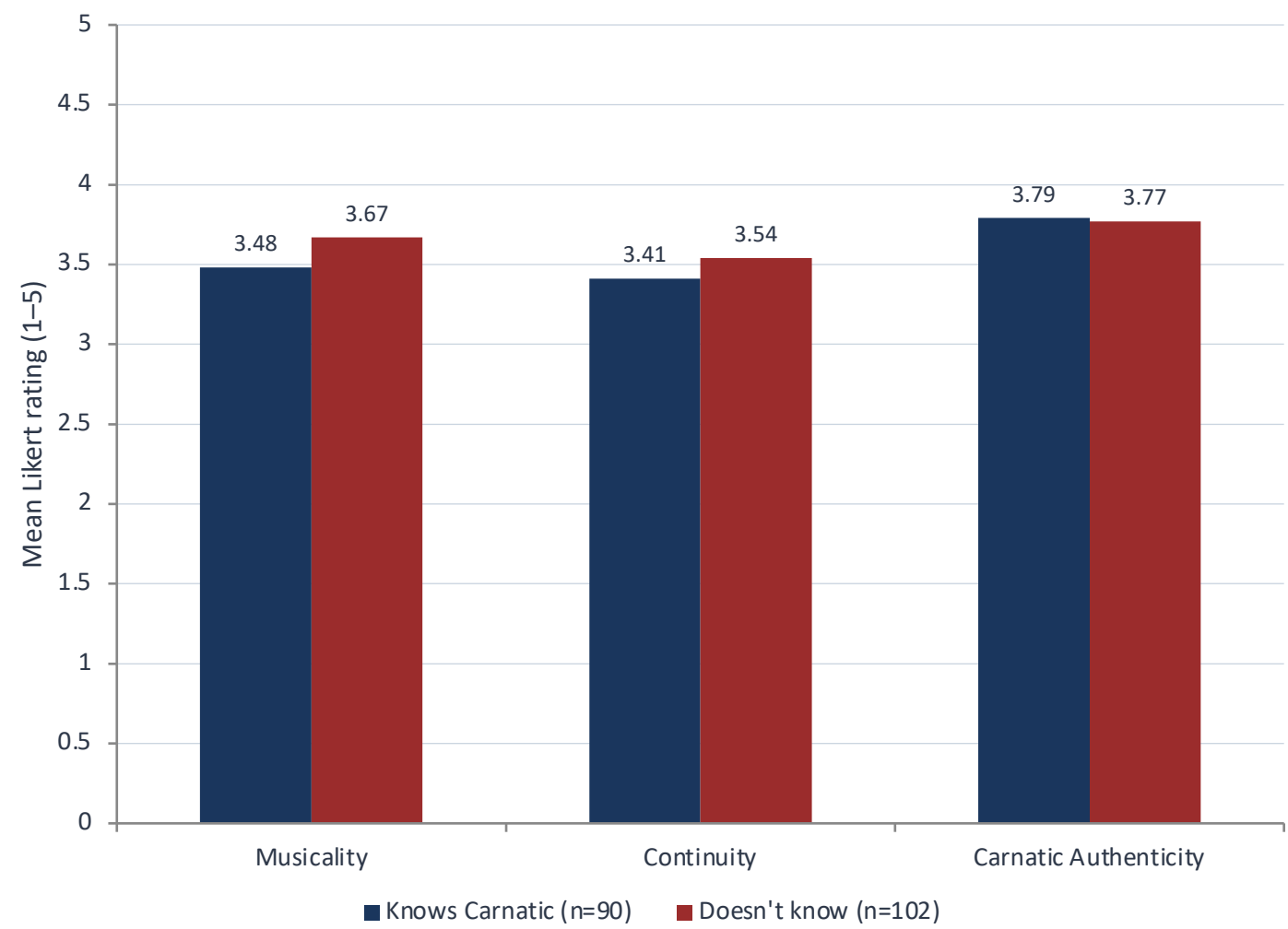
Streamlit web survey, hashed IDs, randomized A/B order — 32 participants, 192 valid trials per task

Participant profile	MOS rating — 6 trials	Pairwise A/B — 6 trials
- Carnatic familiarity (Yes/No) - Formal training level (Beginner → Professional) - Self-rated knowledge (1–5) - Two explicit consent checkboxes	- One 20-s fine-tuned continuation per trial - Balanced across vocal / violin / percussion - Three 5-pt Likert scales: - Musicality · Continuity · Carnatic authenticity	- Two unlabeled 20-s clips per trial (same prompt) - Order randomized per participant - Three forced-choice questions per trial - Free-text comments captured

Participant pool				
32	15	17	192	192
total participants	Carnatic-familiar	Carnatic-unfamiliar	valid Likert trials	valid pairwise trials

Mean Opinion Scores

Fine-tuned continuations rated 3.48–3.78 on 5-pt Likert; Carnatic Authenticity is the highest dimension



Key insight

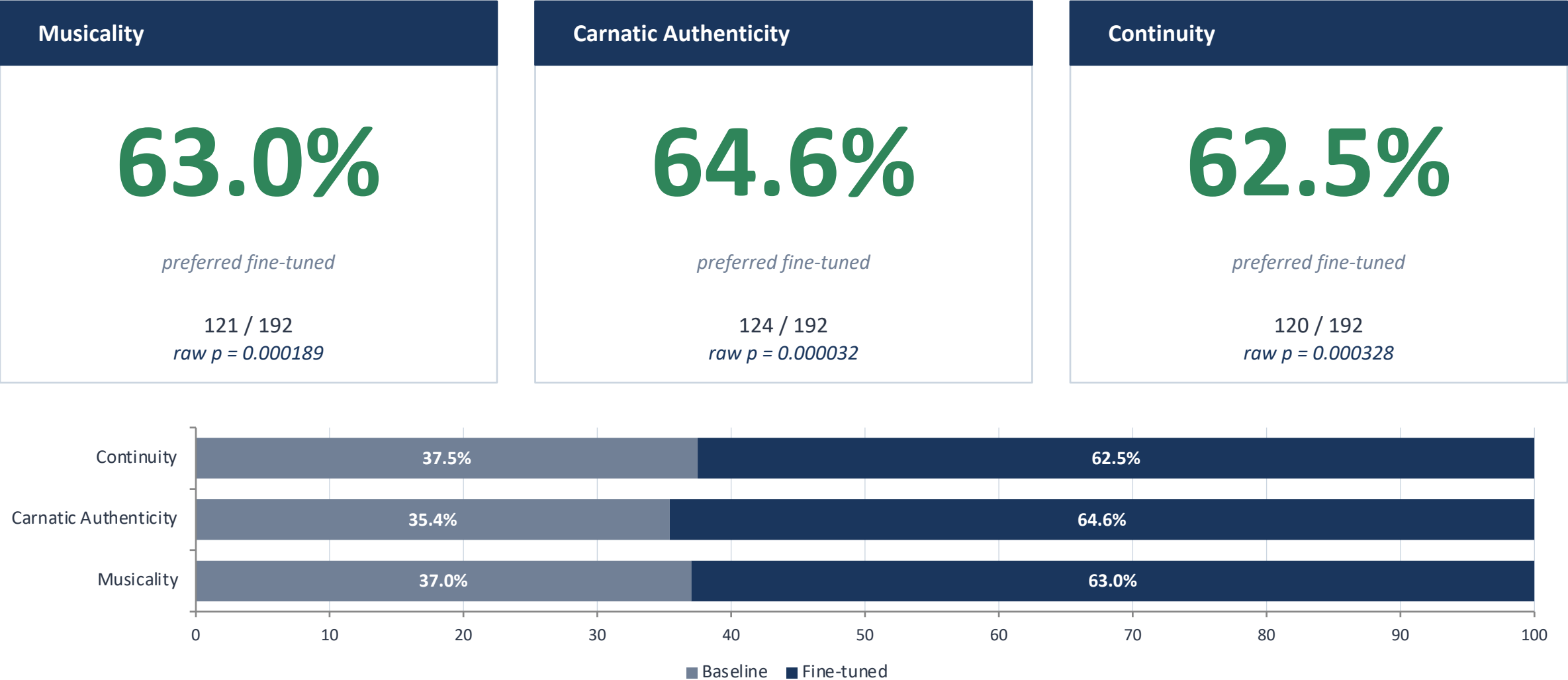
Carnatic-familiar listeners give **lower** scores on Musicality (3.48 vs 3.67) and Continuity (3.41 vs 3.54) displaying a stricter standard.

But on Carnatic Authenticity the two groups agree (3.79 vs 3.77).

Insight: informed listeners hear shortcomings on general musical quality that unfamiliar listeners miss. Yet still recognize the Carnatic stylistic markers the adapter learned.

Pairwise A/B Preferences

Across 192 forced-choice trials per question, fine-tuned wins 62.5%–64.6%



Statistical Testing: Why Holm-Corrected Binomial?

Three small but deliberate decisions in test design

Why a one-sided binomial test

The hypothesis is directional: H_1 asserts fine-tuned > baseline. A two-sided test would dilute power and fail to map onto the actual research claim. The pairwise data is also genuinely binary (clip A vs clip B per question), so binomial is the natural likelihood.

Why Holm correction (not Bonferroni, not raw)

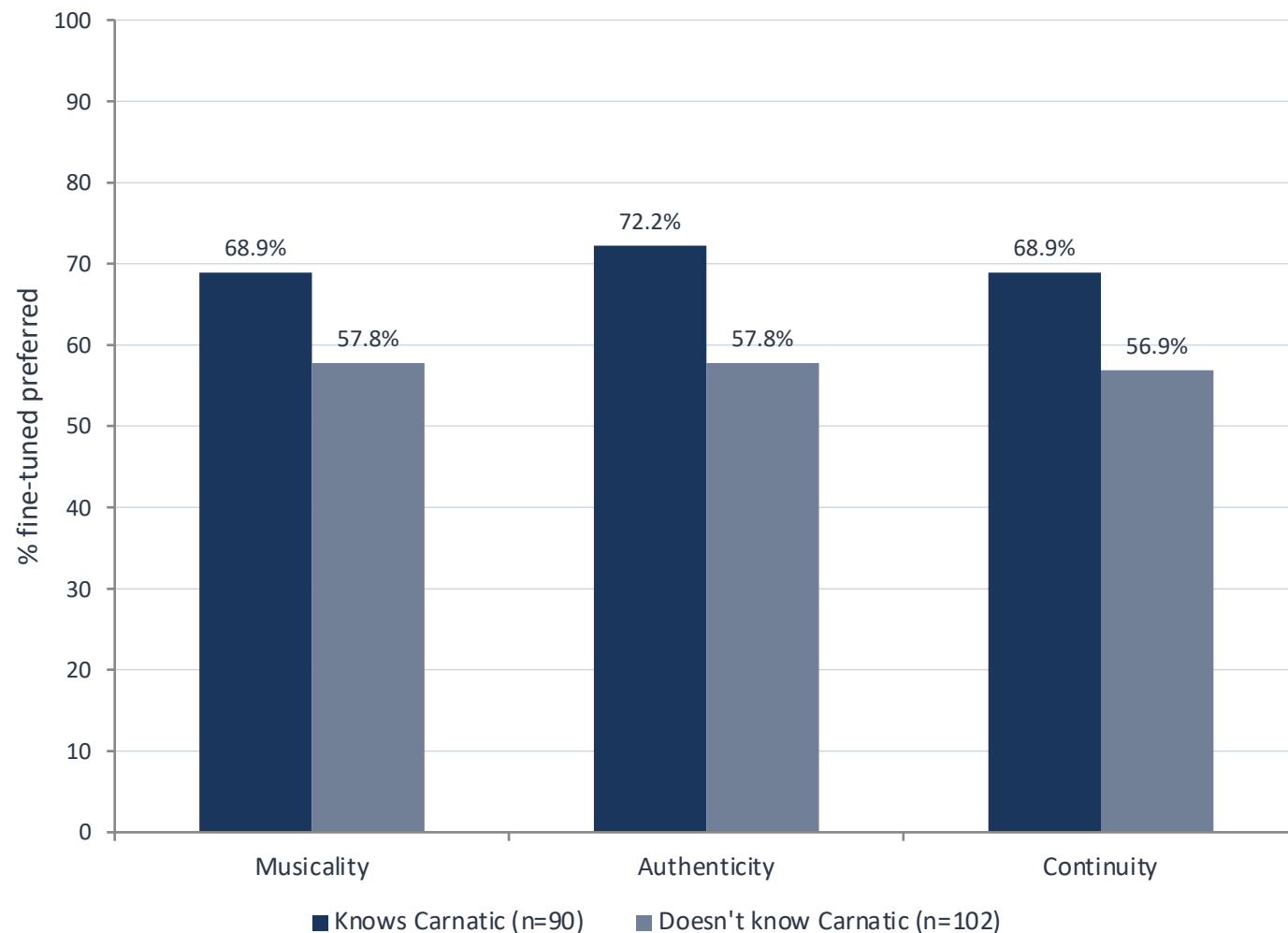
Three pairwise outcomes per analysis (Musicality, Authenticity, Continuity). Bonferroni controls family-wise error rate but is overly conservative. Holm is uniformly more powerful while still controlling FWER at $\alpha = 0.05$. Reporting raw p-values would inflate false-positive risk across three correlated outcomes.

Why $\alpha = 0.05$

Conventional threshold for perceptual studies: keeps results comparable to MusicGen, AudioLM, MusicLM benchmarks. Subgroup analyses (listener category, performance type) apply Holm correction within each subgroup independently.

Subgroup Analysis: Listener Category

The aggregate effect is driven by Carnatic-familiar listeners



After Holm correction ($\alpha = 0.05$)

Knows Carnatic

All three outcomes **significant**

Authenticity Holm $p \approx 4.4 \times 10^{-5}$

Musicality / Continuity Holm $p \approx 4.4 \times 10^{-4}$

H_0 decisively rejected on every question

Doesn't know Carnatic

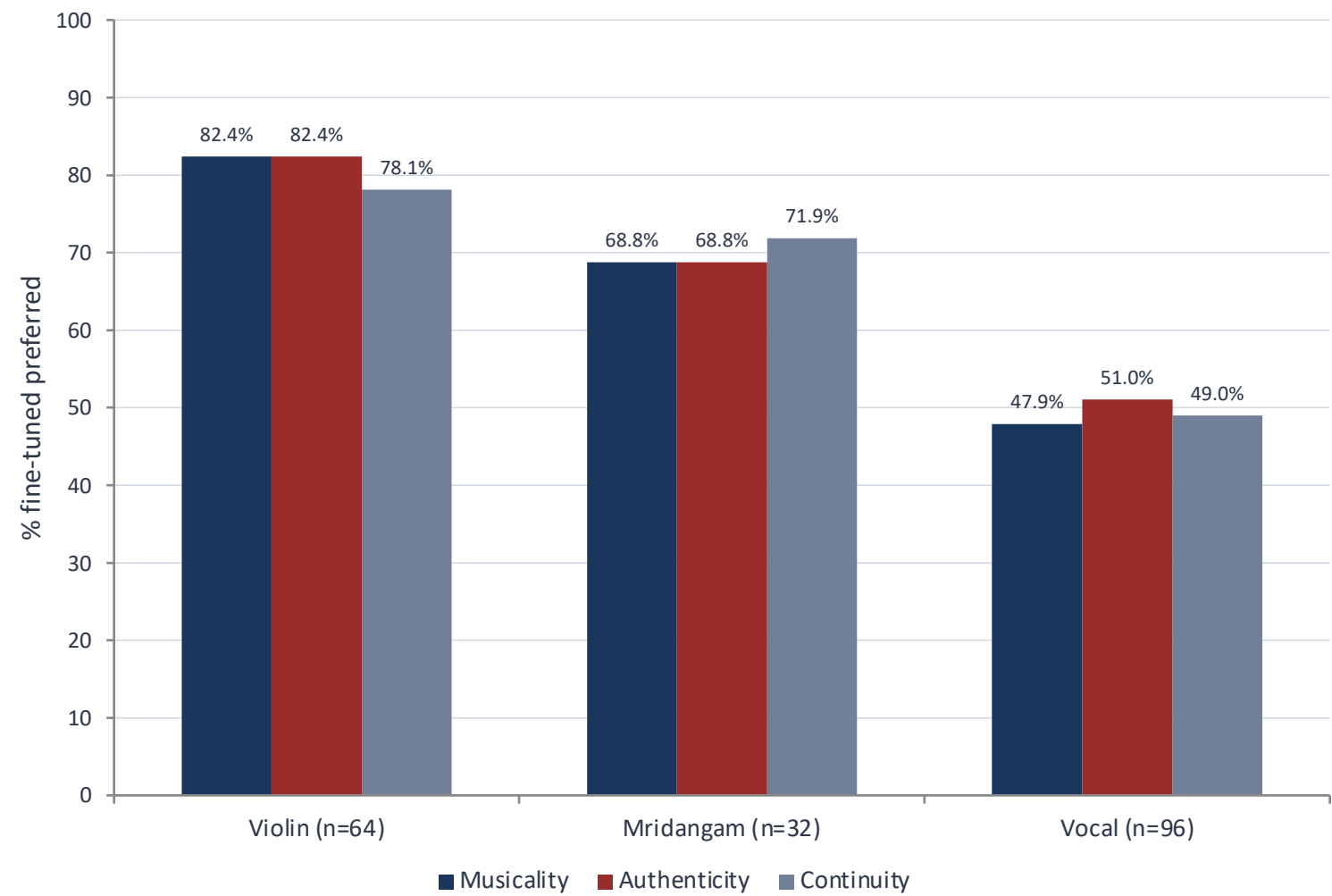
Direction is consistent (~57%)

But Holm-corrected $p \approx 0.21$ — **not significant**

Reading: the adapter is producing culturally-specific changes informed listeners detect

Subgroup Analysis: Performance Type

The fine-tune helps most where the baseline started furthest from authentic Carnatic style



Violin: strongest case

All three Holm-corrected $p < 7 \times 10^{-6}$. Long pitch trajectories and gamakas are exactly where the Western baseline struggles.

Mridangam: directionally clear

69–72% preference, but $n = 32$ + Holm $\times 3$ lifts corrected p past α . Real effect, under-powered.

Vocal: no detectable preference

Hovers at 48–52%. Baseline plausible enough that fine-tune doesn't dominate. Future target.

Synthesis: Answering the Research Questions

Quantitative and perceptual evidence converge

RQ1

Confirmed

Validation CE drops monotonically 4.192 → 4.140 across 10 epochs (perplexity 66.18 → 62.84). ~0.5% trainable parameters shifted the model meaningfully without disturbing the pretrained foundation.

RQ2

Partially confirmed

Mean reductions on Mel (+1.3%), MFCC (+4.2%), Onset (+1.2%); Chroma effectively flat (−0.4%). The largest gain on MFCC matches the timbral / articulation hypothesis. Per-clip wins ~ 50% — noise at clip resolution.

RQ3

Strongly confirmed

All three pairwise preferences significant after Holm correction (62.5%–64.6%, Holm $p \leq 3.8 \times 10^{-4}$). Effect concentrated in Carnatic-familiar listeners and on violin / mridangam — culturally-specific, not generic acoustic.

Headline finding: Modest objective gains + larger perceptual gains + listener-category asymmetry together indicate the adapter is shifting culturally-salient features that fixed-window cosine distances don't directly measure.

Limitation

Five constraints that survived the pipeline; each translates into a future-work direction

1

Canonical FAD not possible

Cleaning pipeline trimmed silence and split into independent 10-s clips before prompts were chosen — no aligned ground-truth continuation tails remain for distributional metrics. Reported boundary cosine distance + Holm-corrected human preferences instead.

2

Per-clip boundary-metric wins $\approx$ 50%

Mean distances improve; per-sample 1-s cosine distance is too noisy a proxy. Stronger metrics (multi-scale spectrograms, learned embeddings, raga-aware pitch metrics) would corroborate at clip resolution.

3

Compute and model-size ceiling

Single Apple Silicon GPU via MPS — restricted to MusicGen-Small. Medium / Large with larger LoRA rank, longer training, mixed precision on CUDA/TPU would likely yield more representational capacity.

4

EnCodec smooths microtonality

MusicGen's tokenizer was trained predominantly on Western audio. Gamaka pitch movements may be partially smoothed by RVQ quantization. A domain-aware tokenizer (or codebook fine-tune) is the natural fix.

5

Listening study scope + horizon

32 participants, 10-second continuation horizon. Long-form Carnatic structure (alapana $\rightarrow$ kriti $\rightarrow$ tani) is outside scope. Larger cohort with more advanced / professional musicians would enable richer subgroup inference.

Future Work

Six directions, prioritized by leverage on the perceptual gap

Re-segment with prompt → reference-continuation pairs

Restores canonical FAD, KL-divergence, embedding-similarity metrics alongside boundary distances.

Scale to MusicGen-Medium / Large on CUDA / TPU

Larger LoRA rank, longer training, mixed precision — compare adapter quality across model sizes.

Fine-tune EnCodec on Carnatic audio

Codebook-only retrain or full encoder/quantizer/decoder co-training so the input tokens themselves become Carnatic-aware.

Carnatic-aware objective metrics

Pitch-class similarity over cents (mod 1200), gamaka extent / modulation index, raga-conditional embedding distances, multi-scale spectral metrics.

Long-horizon continuation (60s, minute-scale)

Evaluate preservation of alapana, kriti development, tala cycles — where structural Carnatic concerns surface.

RLHF + prompt-tuning hybrids

Train a Carnatic preference model from listening-study pairwise judgments (MusicRL-style); combine with LoRA via a learned soft-prompt prefix.

Thank you.

Sundar Ram Subramanian

Code: github.com/sundarram1608/finetuning-musicgen-small-carnatic-continuation

Listening study: capstone-user-evaluation-survey.streamlit.app